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How future work self salience shapes the effects of interacting with artificial intelligence

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ABSTRACT

The rapid rise of artificial intelligence (AI) is transforming the world of work, leaving individuals wondering what AI means for the future of their career. The current research investigates the moderating role of future work self salience (FWSS) on the effect of interacting with AI on perceived control over one's future work self and proactive career behavior. In a first longitudinal experiment with full-time employees in the UK ($N = 174$), participants interacting with AI to solve a task (compared to a control group) experienced increased perceived control over their future work self when FWSS was high, in contrast to those with low FWSS. We replicated this pattern in a second longitudinal study with German business students ($N = 208$). Building on these findings, a third longitudinal experiment with German full-time employees ($N = 155$) extended the model by demonstrating a moderated mediation: for individuals with high FWSS, AI interaction increased perceived control over the future work self and thus promoted proactive career behavior. In contrast, perceived control and proactive career behavior decreased for those with low FWSS. This research demonstrates the potential impact of AI interactions on work-related outcomes, offering critical insights for both theory and practice.

1. Introduction

Recent years have witnessed remarkable advancements in the realm of artificial intelligence (AI), a technology touted to bear significant and perhaps "jarring effects on work and jobs" (Phan et al., 2017, p. 253). The emergence of conversational AI systems like OpenAI's ChatGPT, Google's Bard, and Microsoft's Bing has made this revolution increasingly noticeable, prompting a complex discourse on the impact of AI on our work life and career (Kessler, 2023). Early reports on how AI will affect the world of work both praise the extraordinary potential of AI to boost the global economy (Chui et al., 2023) while also cautioning that AI could affect an estimated 80 % of workers and potentially displace a quarter of the workforce (Eloundou et al., 2023). The impact of AI on individuals' careers is thus likely to be significant (Bankins et al., 2024; Donald et al., 2024). While AI is likely to create new roles and even new industries, it will also fundamentally change or even replace existing jobs, requiring individuals to develop new skills and making at least some of their extant competencies redundant (Behrend et al., 2024; Selenko et al., 2022). The resulting ongoing need to continually update skills and knowledge means that individuals' proactive efforts to develop their skills and shape their career is likely to play an increasingly significant role (Hirschi, 2018; Lent et al., 2022).

However, previous research suggests that individual perceptions of AI vary widely (Cave & Dihal, 2019), particularly with regard to

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its integration into the workplace (Bankins et al., 2023; Selenko et al., 2022). Thus, how individuals react to the impact of AI depends not only on "the type of technology implemented," but also on "individual differences among workers" (Bankins et al., 2023, p. 11; see also Lyndgaard et al., 2024). While some initial studies have begun to explore how individual-level factors influence responses to AI, research has primarily focused on how personality influences attitudes towards AI (Kaya et al., 2024; Stein et al., 2024) and on how individuals' feel their current job or employment prospects are affected by AI (Bhargava et al., 2021; Lin et al., 2024). How individuals respond to interactions with AI is however profoundly affected by their perceptions of what these experiences might mean for their future (Gioia & Thomas, 1996; Gioia et al., 1994). Yet, how individuals' more future-focused career-related cognitions shape interactions with AI and thus their career-related behaviors remains largely unknown.

To address this issue, we draw on the proactive motivation model (S. K. Parker et al., 2010) as well as on research on future work selves, i.e., individuals' cognitive representations of their future in relation to work (Strauss et al., 2012). The proactive motivation model suggests that the interplay between "reason to" as well as "can do" factors shapes individuals' motivation for proactive behavior. Drawing on this overarching theoretical framework, we propose that interacting with artificial intelligence and witnessing its capabilities will affect individuals' sense of control over their future work self (a "can do" factor), depending on the initial salience of their future work self (a "reason to" factor). Future work self salience (FWSS) reflects how "clear and easy to imagine" the future work self is for the individual (Strauss et al., 2012, p. 581). We argue that FWSS will buffer the impact of AI on individuals' sense of control over their future: on the one hand, having a clear image of their future work self will allow individuals to envision the benefits of AI for their future career, leaving them feeling more in control of their desired future and thus promoting their proactive career behavior. On the other hand, those with low levels of FWSS will experience a lack of control over their future when they interact with AI, making it less likely for them to engage in proactive career behavior. Fig. 1 shows our theoretical model.

Our study contributes to the literature in the following ways. First, we provide insights into the role of AI for individuals' future-oriented career-related cognition and behavior. Previous studies have predominantly adopted a "present" perspective when investigating employees' views of AI (e.g., Bochniarz et al., 2022; Kong et al., 2023; Langer et al., 2023), for example, examining their beliefs about the impact of AI on their current employability and job security in their existing job (Bhargava et al., 2021; Lin et al., 2024). However, the notion of prospective sensemaking (Gioia et al., 1994; Gioia & Thomas, 1996) suggests that responses to current experiences, such as AI interactions, are also strongly shaped by individuals' conceptions of what this might mean for them in the future. By emphasizing how future selves act as a "homing beacon," (Ashforth & Schinoff, 2016, p. 115), our study demonstrates that individuals interpret AI interactions not only in terms of immediate challenges and opportunities, but also in terms of how these interactions align with and inform their future career. We thus add a novel perspective to emerging research (Tang et al., 2023) that emphasizes that AI interactions have the potential to lead to both positive and negative outcomes (i.e., while some individuals may feel more control over their future when interacting with an AI, others may experience the opposite). As a result, our research provides a more comprehensive view of how work-related interactions with AI will impact employees.

Second, we expand theory on future work selves by introducing perceived control over the future work self as an important aspect of career-related future-oriented cognition. To date, research on future work selves has predominantly focused on their salience, and numerous studies have shown that FWSS promotes individuals' efforts to proactively shape their careers (Guan et al., 2017; Strauss et al., 2012; Taber & Blankemeyer, 2015). We introduce individuals' perceived control over their future work self, as an important yet previously overlooked aspect of future work selves, reflecting "can do" motivation. Future work selves are a specific type of possible self (Strauss et al., 2012), and in the wider literature on possible selves, the amount of control an individual feels they have over their possible self has long been considered an important factor that determines individuals' efforts to bring about this possible self. For example, Oyserman and James (2009) highlight that the perceived controllability of a possible self determines the effort individuals

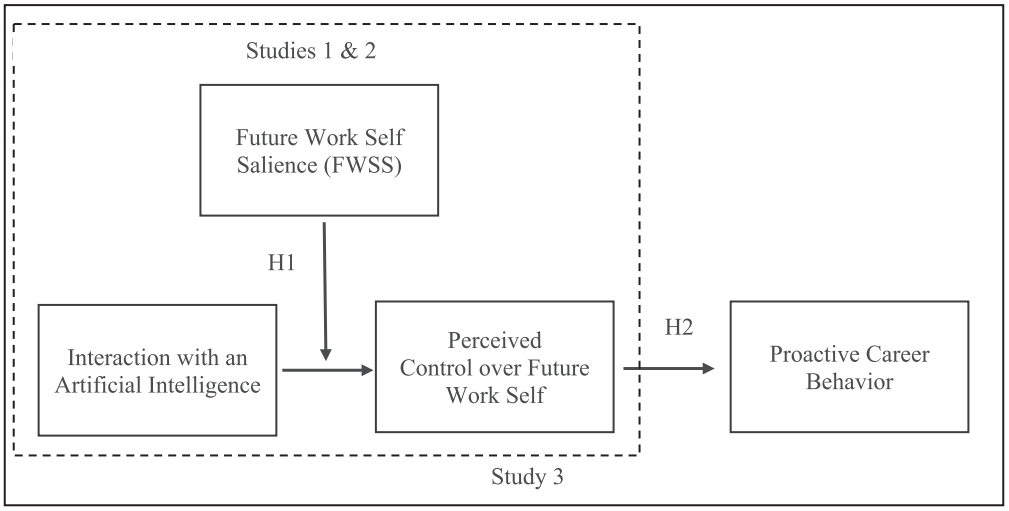


Fig. 1. Theoretical Model.

invest in realizing it. Yet, in relation to future work selves, individuals' sense that the attainment of their future work self is in their control has not been considered.

In addition, we extend research on FWSS by investigating it as a boundary condition. Studies to date have predominantly focused on the main effects of FWSS on career-related outcomes. However, some researchers have instead considered how FWSS can shape employees' reactions to their environment. For example, Yu et al. (2016) found that employees high in FWSS reported lower levels of affective commitment when their supervisor was abusive. Xu et al. (2021) found that for healthcare workers high in FWSS, supervisor career support had a less pronounced effect on career commitment. Lin et al. (2024) found that employees high in FWSS reported more proactive career behavior when their self-perceived employability was low. We contribute to this nascent stream of research, which shows that FWSS can shape individuals' reactions to averse (e.g., abusive supervision or reduced employability), positive (e.g., supervisor support), or ambiguous (e.g., AI) contexts in organizations. This is of particular importance as AI systems are increasingly integrated into human workplaces (Duan et al., 2019; Jarrahi, 2018) in what we may think of as a new industrial revolution (Pereira et al., 2023). Thus, our results offer valuable insights for leaders aiming to comprehend the influence of AI integrations on employees' career-related cognitions and behaviors (Budhwar et al., 2023).

1.1. Literature review and hypotheses

1.1.1. AI and individuals' careers

AI broadly refers to a wide range of computerized systems that "learn from experience, adjust to new inputs, and perform human-like tasks" (Duan et al., 2019, p. 63). As such, AI is an umbrella term that encompasses a range of technologies and tools (e.g., machine learning, deep learning, and natural language processing) enabling task automation, large-scale data analysis, predictive modeling, and the emulation of human decision-making (H. Lu et al., 2018; von Krogh, 2018). Researchers propose that AI has the potential to augment human creativity (Berg et al., 2023; Epstein et al., 2023), boost productivity (Agrawal et al., 2023), and improve overall performance (Brynjolfsson et al., 2023). Consequently, integrating AI into the workplace is poised to profoundly affect the nature and trajectory of individuals' careers (Bankins et al., 2024; Gati & Kulcsár, 2021; Kong et al., 2023; S. K. Parker & Grote, 2022). As described by Donald et al. (2024, p. 3), AI is a "disruptive force with a high potential to change the shape of the labor market and career ecosystem," posing both a threat and "new opportunities to careers in all industrial sectors".

Yet, the relationship between AI and individuals' perception of its influence on their careers is complex. Importantly, AI systems have the capacity to both automate and augment tasks – the former suggesting machine replacement of human labor, the latter indicating collaborative efforts between humans and machines (Raisch & Krakowski, 2021). While the "type of technology implemented" is likely to influence the automating vs. augmenting effect of the AI on labor, individual reactions to the impact of AI are "multifaceted" and also depend "on individual differences among workers" (Bankins et al., 2023, p. 11, see also Maragno et al., 2023). Thus, individuals differ in how they feel about the rise of AI (Kaya et al., 2024; Stein et al., 2024), and thus how they react to it.

For some individuals, encountering AI's capabilities may be associated with uncertainty about their future career, triggering feelings of a loss of control over their aspired-to future and negatively impacting their efforts to prepare for the future. In contrast, other individuals may interpret the same interaction as increasing their level of control over their professional future, viewing AI as an instrumental tool that empowers them to actively shape their career. By adopting a future-oriented perspective, we extend the current perspectives on individuals' present views towards AI and its impact on careers. Specifically, we examine how individuals' interactions with AI, influenced by their perceptions of what it might mean for them in the future, may alter their sense of control over their future career. Below, we describe how encountering AI and its capabilities will interact with individuals' future work selves, specifically, how future work selves are impacted by AI and how they, in turn, shape individuals' reactions to it.

1.1.2. Future work selves as a lens for understanding reactions to AI

The rise of AI may affect individuals' career-related cognition and behavior in a multitude of ways; for example, it may trigger concerns about employability (Yam et al., 2022) and uncertainty about the future job market (Frank et al., 2019). We expand current work on individuals' reactions to AI by adopting a future-oriented perspective. In particular, we draw on the literature on prospective sensemaking (Gioia et al., 1994; Gioia & Thomas, 1996), which suggests that reactions to current experiences, such as interactions with AI, are also strongly shaped by individuals' conceptions of what this might mean for them in the future. Building on this framework, we suggest that the impact of AI on future-oriented career-related cognitions and behaviors can be best understood through the lens of future work selves. Future work selves are individuals' representations of who they want to become in the future, reflecting their hopes and aspirations regarding work (Strauss et al., 2012). They are a type of possible self (Markus & Nurius, 1986), reflecting individuals' cognitive manifestation of who they want to become at work. Future work selves are related to other concepts in the career literature such as occupational identity, "the clear perception of occupational interests, abilities, goals, and values" (Hirschi, 2012, p. 480), career aspirations, the desire to advance in one's chosen career field (O'Brien, 1996), or career commitment, "the development of personal goals, the attachment to, identification with, and involvement in those goals" (Colarelli & Bishop, 1990, p. 159). Yet, future work selves are a distinct concept: they form part of the self-concept and are thus explicitly identity-oriented. As such, they "function as the personalized carriers (representations) of general aspirations, motives, and threats" (Markus & Nurius, 1986, p. 955). A focus on future work selves thus allows us to investigate the impact of AI on career-related threats and aspirations as they are embedded in the self. Future work selves make the future of work "personal" and relevant to individuals' identity, in contrast to more general beliefs such as uncertainty about the future labor market (e.g., "Will my planned occupation [e.g., accountant] disappear due to technological changes?"; (Gati & Kulcsár, 2021, p. 9) or perceived employability (e.g., will I be able to keep my job or get one I want; Rothwell & Arnold, 2007).

We propose that experiencing the powerful capabilities of an AI will affect the amount of control individuals feel they have over the attainment of their future work self. Norman and Aron (2003) defined perceived control over a possible self as the extent to which its attainment is perceived as under one's control and showed its importance for individuals' efforts to achieve the respective possible self. Future work self control thus reflects individuals' control appraisal in relation to their future self, i.e., their perceived control over their future in relation to work considering their available resources (Lazarus & Folkman, 1984).

1.1.3. The moderating effect of future work self salience

A key feature that determines the motivational impact of a future work self is its salience: The degree to which the future work self is"clear and easy to imagine" (Strauss et al., 2012, p. 581). Strauss et al. (2012) suggest that the motivational potency of a salient future work self can be explained through episodic prospection, i.e., the mental simulation of the future (Bulley et al., 2016; Suddendorf & Moore, 2011). This mental simulation of possible future scenarios allows for the organization of current actions in anticipation of future events, consistent with the principle that imagining future states optimizes goal-directed cognition and behavior (Seligman et al., 2013). Thus, by mentally simulating their future work selves, individuals can identify potential discrepancies between their current abilities and future demands (Strauss et al., 2012), which in turn motivates action to bridge these discrepancies (Schultz & Hernes, 2013).

According to Strauss et al. (2012), this mental simulation of the future allows individuals to identify obstacles as well as opportunities in relation to their future self. In support of this, Taylor et al. (1998) demonstrated that mentally simulating possible future events facilitates planning and problem solving (for a recent meta-analysis see Cole et al., 2021). Furthermore, simulating stressful events has been shown to increase the propensity to engage in problem-solving activities (Jing et al., 2016; Rivkin & Taylor, 1999). During periods of uncertainty, individuals with salient hopeful possible selves experience greater psychological adjustment and exhibit lower levels of anxiety and depression (Sweeny & Dunlop, 2020), highlighting the stabilizing effects of a well-defined future work self. Social cognitive theory also posits that mental simulation provides a vicarious experience that increases confidence in achieving desired outcomes (Bandura, 1989). In summary, overall there is substantial evidence that mental time travel promotes motivation and goal pursuit (Baumeister et al., 2016). Consequently, we argue that individuals with a salient future work self are able to mentally travel into the future and imagine possible uses of AI in their future working lives. Rather than seeing AI as a threat to their future work self, FWSS allows them to imagine its possible benefits, as well as ways around its potential negative impact, leaving individuals feeling more in control of their future work self. In contrast, when FWSS is low, individuals are likely to struggle to envision their future in relation to work (Strauss et al., 2012), providing them with limited opportunities to imagine possible advantages of AI for their future work. Instead, AI may seem as limiting their future opportunities, resulting in a perceived lack of control over the future self. Thus, we propose:

Hypothesis 1. FWSS moderates the relationship between interaction with an AI (compared to a control group) and future work self control, such that the relationship is positive when individuals' FWSS is high and negative when individuals' FWSS is low.

1.1.4. Effects on proactive career behavior

Future work selves have important self-regulatory functions in the context of careers (Fang & Saks, 2022) and motivate individuals' efforts to bring about their desired future in relation to work through proactive career behavior (Han & Hwang, 2022; Strauss et al., 2012; Taber & Blankemeyer, 2015). As S. K. Parker et al. (2010) argued, a salient future work self can provide "reason to" motivation for proactive career behavior. Proactive career behavior is defined as the self-initiated actions individuals undertake to shape their future career, such as setting goals, acquiring skills, and building professional relationships (Claes & Ruiz-Quintanilla, 1998; Strauss et al., 2012). Proactive career behavior has been proposed to comprise four dimensions: career planning, career consultation, skill development, and networking (Claes & Ruiz-Quintanilla, 1998; Strauss et al., 2012). Hence, proactive career behavior is an umbrella term that includes various career-related activities and behaviors, including career self-management, a specific form of proactive career behavior (Hirschi & Koen, 2021).

Prior research that has linked future work selves to proactive career behavior has focused on FWSS, drawing on Strauss et al. (2012), essentially concentrating on the "reason to" motivational aspect of future work selves. However, individuals' "can do" motivation in relation to their future work self, their perceived control, has received little attention. In the wider literature on possible selves, however, Norman and Aaron have shown that "the extent to which the attainment [of a possible self] is perceived as under one's control" (p. 500) predicts individuals' motivation to attain it. In their review of the possible selves literature, Oyserman and James (2009) emphasize that the perceived controllability of a possible self determines the effort individuals invest in bringing it about. This is because control- or expectancy beliefs, individuals' expectations that their efforts can produce the desired outcome, determine their choices and behavior (Bandura, 1997; Eccles & Wigfield, 2002). In further support for this notion, Koen and Parker (2020) found, in a sample of workers in unstable jobs, a negative relationship between workers' perceived control (at work) and their proactive career behaviors. There is thus theoretical and empirical support for the argument that perceived control over the future self, in conjunction with FWSS, is likely to promote individuals' efforts to shape their career and that the interplay between future work self control and FWSS determines the impact of AI on proactive career behavior. When FWSS is high and individuals feel in control of their future, this provides them with reason-to and can-do motivation to take charge of their future career. In contrast, when FWSS is low, individuals will lack a guiding image of the future and thus lack the "reason" to engage in proactive career behavior. In sum, we propose:

Hypothesis 2. Individuals' FWSS moderates the indirect relationship between interaction with an AI (compared to a control group) and proactive career behavior through future work self control, such that the indirect effect is positive when individuals' FWSS is high and negative when individuals' FWSS is low.

2. Overview of studies

We tested our hypotheses in three longitudinal online experiments in which participants interacted with the generative AI ChatGPT. While a myriad of AI technologies have already been incorporated into workplaces (von Krogh, 2018), the spotlight has recently shifted to generative AI systems (i.e., AI systems that "leverage deep learning models to generate human-like content (e.g., images, words) in response to complex and varied prompts (e.g., languages, instructions, questions", Lim et al., 2023, p. 2). Because of their potential to impact a wide variety of professions that have traditionally been seen as relatively unaffected by automation (Dwivedi et al., 2023), generative AI systems are a pertinent context for the study of the impact of AI on individuals' career-oriented cognition and behavior. Addressing recent calls to enhance the realism of AI-related studies (Tang et al., 2023), our research involved genuine interactions with ChatGPT rather than merely asking individuals to imagine interacting with an AI. By developing a cooperation task with ChatGPT, we aimed to capture the nuances of real work experiences with AI (Tang et al., 2023). In Study 1, we investigated the moderating effect of FWSS on the relationship between experimental condition (interacting with an AI vs. control group) and future work self control at Time 1 and one week later at Time 2 in a sample of working adults in the UK who reported never having interacted with ChatGPT before. In Study 2, we used the same experimental design in a sample of business students in Germany. In Study 3, we used this experimental design to investigate the full moderated mediation model in a sample of full-time employees in Germany. The data of all studies, analyses code, and study materials are available at the Open Science Framework: https://osf.io/wpdy5/?view_only=316f232b7f4f49ea9e18fae7b58b9bc6.

3. Study 1

3.1. Method

3.1.1. Sample and procedure

The study was approved by the research ethics committee of the second author's institution ("ChatGPT and future work selves," Ref: 2023–05). We recruited full-time employees in the UK via Prolific Academic (Palan & Schitter, 2018). We first invited people to participate in a pre-screening survey using demographic filters that met the following criteria: Individuals who were employed full-time in the UK, fluent in English, and with a minimum of 90 % approval rating on the platform. Approximately 1000 pre-screening surveys were sent out (with a payment of £0.10, US$0.13). Participants answered the question if they had ever used the chatbot *ChatGPT*.[1] 402 participants (40 %) reported having used ChatGPT, and 608 participants (60 %) reported not having used ChatGPT. Of the latter, we randomly selected 226 participants and invited them to participate in the experiment, which consisted of two surveys one week apart. Two hundred nineteen participants completed the Time 1 survey, and 179 completed the Time 2 survey. Participants received financial compensation of £4.75 (US$6.00) for the Time 1 survey participation and £0.75 (US$0.95) for the Time 2 survey.

Adhering to state-of-the-art guidelines (Ward & Meade, 2022; Zickar & Keith, 2023), we checked for careless responses using instructed response items (Meade & Craig, 2012) and for multivariate outliers applying Mahalanobis distance (cutoff p-value of 0.001, Tabachnick & Fidell, 2013). Five participants who failed more than one attention check were excluded, while no exclusions were made based on Mahalanobis distance. Therefore, our final sample consisted of 174 participants (36.8 % identified as women, the remaining participants identified as men; age: $M = 42.16$ years, $SD = 9.94$).

Utilizing Strauss et al.'s (2012) procedure, participants were directed to mentally project themselves into the future and imagine the future work self they hoped to achieve. Following this, participants rated the salience of their future work self. Then, participants worked on a structured in-tray task (adapted by Chernikova et al., 2016; S. L. Parker et al., 2009, 2013). They were prompted to take on the role of Interim Retail Manager Alex Jennings at a struggling fictional Borough Bank branch, tasked with improving the branch's performance. The structured in-tray task consisted of three stages, each corresponding to a specific document: sales figures, survey results, and a customer complaint letter. For each document, participants were required to (1) draft a list of actions based on their analysis of the document's key issues, (2) assign priority to these actions, and (3) suggest individuals to execute these actions.

All participants initially worked independently on the first document, the sales figures. Participants wrote their answers to the three tasks in three free-text fields. After submitting their responses to the first document, participants were randomly allocated to one of two conditions.

In the *AI-assisted condition*, participants were informed that an AI, ChatGPT, would support them with their upcoming tasks. Specifically, a separate browser window opened automatically for these participants, leading to the ChatGPT website.

As participants started working on the second document, they were prompted to input pre-constructed instructions into ChatGPT, entailing necessary background details about Alex Jennings, his role, and the organizational structure. Subsequently, they entered pre-formulated instructions (prompts)[2] into ChatGPT for each task and recorded ChatGPT's responses to each task in three free-text fields.

[1] We recruited participants who had never used ChatGPT to capture their initial reactions and interactions. This approach provided a unique opportunity to observe unbiased first impressions and the immediate impact of interacting with AI-powered conversational agents.

[2] We used pre-written prompts for the first task with ChatGPT for two main reasons: First, participants in Study 1 were completely unfamiliar with the technology. Given that inexperienced users often struggle to effectively use large language models due to lack of familiarity and training (Bašić et al., 2023), which typically leads to a trial-and-error process of developing prompts (Zamfirescu-Pereira et al., 2023), this approach was intended to simplify the initial interaction with the AI. Second, this structured process gave us more control over what happened between the participants and the AI, making it more likely that all participants would have a relatively similar experience.

After submitting their responses, they moved on to the third document. Participants were again reminded that they would maintain their roles as Alex Jennings and were free to continue utilizing ChatGPT. They were provided with sample instructions for this interaction but encouraged to write their own prompts.

In the *control condition*, participants independently completed the same tasks for the second and third documents without interacting with ChatGPT.

Following this experimental manipulation, participants were reminded of their previously imagined future work self and rated how much control they believed to have over attaining this future work self, using the measure by Norman and Aron (2003).

One week later, participants were invited to the second part of the study. Participants again reported their perceived control over their future work self and how much they had used ChatGPT during the past week. Fig. 2 shows the study design.

3.1.2. Measures

3.1.2.1. Future work self salience (pre-test). Participants' FWSS was assessed prior to the experiment with three items (α = 0.93) by Strauss et al. (2012) (e.g., "I am very clear about who and what I want to become in my future work."; 1 = *strongly disagree*, 7 = *strongly disagree*).

3.1.2.2. Perceived control over the future work self (post-test Time 1 and 2). Participants responded to six items (α = 0.88 and 0.88 for T1 and T2, respectively) developed by Norman and Aron (2003) to assess their perceived control over their future work self (e.g., "How much control do you believe you have over attaining this particular hoped-for future self?"; 1 = *none at all*; 7 = *a great deal*).

Perceived control over the future work self is conceptually distinct, yet related to, concepts such as locus of control (Lefcourt, 1976; Rotter, 1966), career confidence (Hirschi et al., 2017), and career-related self-efficacy (Kossek et al., 1998). In order to establish its divergent and convergent validity in relation to related concepts, we conducted a separate validation study with 257 full-time employees in the UK recruited via Prolific (see Online Supplemental Material for a detailed description of Method and Results). We measured future work self control and FWSS as in Study 1, as well as career self-efficacy (10 items; Kossek et al., 1998), career-related confidence (4 items; Hirschi et al., 2017), job-related control appraisal (4 items; S. K. Parker et al., 2006), participants' perceived control over their life and future (8 items; Infurna et al., 2011), the control dimension of career adaptability (6 items; Savickas & Porfeli, 2012), and internal and external locus of control (4 items; Nießen et al., 2022).

Both the average variance extracted (AVE) and composite reliability (CR) for future work self control surpassed the recommended thresholds of 0.50 (Hair et al., 2017) and 0.70 (Hair et al., 2014), respectively (see Table S1 in the Online Supplemental Material). In order to determine discriminant validity, we considered three metrics. First, the heterotrait-monotrait (HTMT) values were consistent with the conservative 0.85 cutoff proposed by Henseler etal. (2015) (see Table S2 in the Online Supplemental Material). Second, adopting the CI_{CFA} (sys) approach suggested by Rönkkö and Cho (2022), we inspected the 95 % confidence intervals (CIs) for correlations between the future work self control measure and all related constructs. If either the upper or the lower limit of the CI surpasses 0.90, this is considered to indicate a potential issue in validity. No correlation between future work self control and associated constructs crossed this benchmark. Lastly, a CFA model distinguishing between the 9 different concepts significantly outperformed all alternative models (see Table S3 in the Online Supplemental Material). In conclusion, the results of the validation study provide robust support for both the convergent and discriminant validity of future work self control.

3.1.2.3. Control variables. We controlled for participants' age, gender, and education, as prior research has shown that these variables are related to an individual's perception of control over future events (Darvill & Johnson, 1991; Elst et al., 2011; Infurna et al., 2016; Specht et al., 2013). Moreover, we considered self-reported knowledge about AI technology as a possible confounding variable, given that participants who are more knowledgeable about AI technology may better understand how to utilize AI in their work and thus perceive a higher degree of control over their future work self (Said et al., 2023). We used an item developed by Gaube et al. (2021) ("How would you consider your general knowledge of artificial intelligence (AI)?"; 1 = *I have no knowledge*; 2 = *Novice: I have heard of AI;* 3 = *Intermediate: I have read media articles or have listened to news about AI technologies*; 4 = *Advanced: I have used AI-based tools and have some understanding of how they work*; 5 = *Expert: For example, I am an academic or industry researcher in AI*). Lastly, at T2, we assessed ChatGPT usage in the past week ("Since the last survey, how much time did you spend using ChatGPT?"; *1 = none at all, 5 = a great deal*). We assumed that it was possible that participants in the experimental group who interacted with ChatGPT in our study and had not previously used it would continue using it, which could alter their perception of the influence of AI on their work-life (Gansser & Reich, 2021; Kim & Malhotra, 2005; Lu et al., 2019). The results retain the same pattern when these covariates are not included (Bernerth & Aguinis, 2016).

3.2. Results

Descriptive statistics and correlations of study variables can be found in Table 1. We followed Preacher et al. (2007) and Hayes's (2022) recommendations to examine our first hypothesis and used the SPSS PROCESS macro (Model 1, Hayes, 2022). FWSS was mean-centered before creating its interaction term.

First, we examined Hypothesis 1, which predicted that FWSS would moderate the relationship between the experimental condition (coded as 1 = *interacting with an AI*, 0 = *control group*) and future work self control, such that the relationship is positive when individuals' FWSS is high and negative when individuals' FWSS is low. As shown in Table 4 (Study 1), we regressed future work self

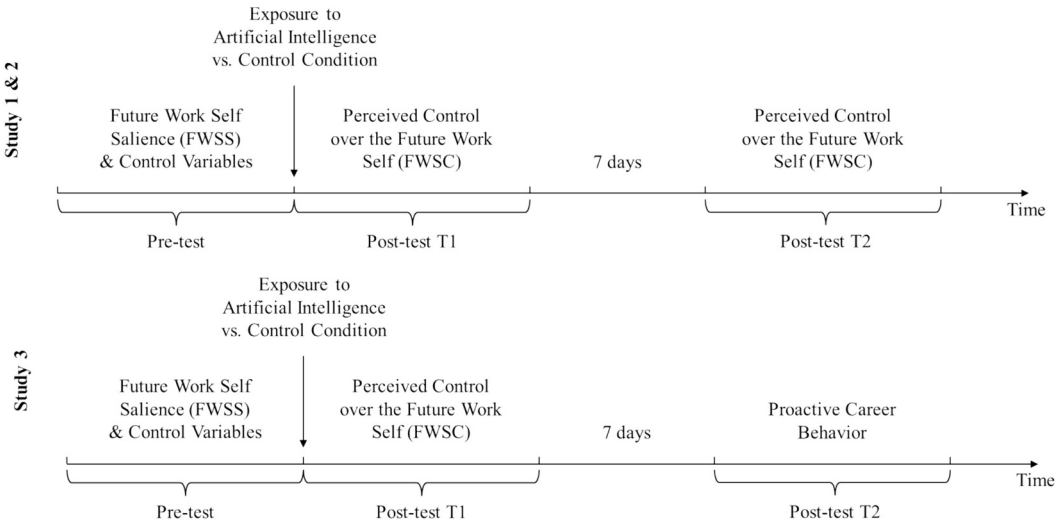


Fig. 2. Schema of the study designs.

control at T1 (Model 1) and at T2 (Model 2) on the experimental condition, FWSS, and the interaction term between experimental condition and FWSS.

The main effect of the experimental condition on perceived control over the future work self at T1 was not significant ($\beta = 0.09$, $p = .43$; Table 4, Study 1, Model 1). However, we found a significant interaction effect between the experimental condition and FWSS on perceived control over the future work self at T1 ($\beta = 0.26$, $p < .05$; Table 4, Study 1, Model 1). Fig. 3 illustrates the pattern of the interaction. A simple slope analysis (Aiken & West, 1991) revealed a positive and significant effect of experimental condition on perceived control over the future work self for individuals with high FWSS (1 *SD* above the mean; slope = 0.35, $p < .05$) while the effect was not significant for those with low FWSS (1 *SD* below the mean; slope = −0.16, $p = .32$). Moreover, these slopes differed significantly from one another, $z = 2.25$, $p < .05$ (Paternoster et al., 1998).

We performed the same analysis for perceived control over the future work self at T2. Again, the main effect of the experimental condition was not significant ($\beta = -0.02$, $p = .90$; Table 4, Study 1, Model 2). However, as expected, the interaction effect between the experimental condition and FWSS on perceived control over the future work self at T2 was significant ($\beta = 0.25$, $p < .05$; Table 4, Study 1, Model 2). We conducted a simple slope analysis which revealed that both slopes were in the expected direction but not significant (high FWSS: 1 *SD* above the mean, slope = 0.23, $p = .16$; low FWSS: 1 *SD* below the mean, slope = −0.26, $p = .12$). The difference between these slopes was significant ($z = 2.18$, $p < .05$).

3.3. Discussion

The findings from Study 1 provided initial support for Hypothesis 1. Specifically, the relationship between interacting with an AI (versus the control condition) and perceived control over the future work self was positive for participants with high FWSS. Individuals with a salient future work self seemed to have a greater sense of control over their future work self after interacting with ChatGPT for the first time, while those with less salient future work selves do not reap the same benefits.

Even though these findings are encouraging, Study 1 has a number of limitations. First, we specifically pre-screened and selected participants who had not previously used ChatGPT in order to rule out that varying levels of prior experience with this AI would affect our results. However, the effects of interacting with ChatGPT on future work self control may be different for those who are already familiar with it, raising concerns about the generalizability of our findings to a population that increasingly uses this AI tool (Wojcieszak et al., 2021). In addition, the effects of FWSS may differ for groups in which concerns about their future career are more salient than they likely are for our sample of full-time employees. To address these constraints, we conducted Study 2 with a sample of university students with varying levels of prior ChatGPT use.

4. Study 2

4.1. Method

4.1.1. Sample and procedure

Participants were students at a large German business school. A total of 243 students participated in the first part of the experiment and 231 in the second part. We eliminated five incomplete and duplicate surveys and six participants who could not be matched between the two-time points. Moreover, following the same approach as in Study 1, we followed recent guidelines (Ward & Meade, 2022; Zickar & Keith, 2023) and checked for careless responses using instructed response items (Meade & Craig, 2012), and for

Table 1
Descriptive statistics and correlations between study variables (Study 1).

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Condition	0.49	0.50	–													
2. FWSS	3.47	1.04	−0.01	–												
3. Perceived control over the FWS T1	5.46	0.95	0.03	0.54**	–											
4. Perceived control over the FWS T2	3.35	0.79	−0.00	0.54**	0.80**	–										
5. Age (years)	32.2	9.16	−0.14	0.19*	0.06	0.06	–									
6. Gender	0.37	0.48	−0.13	−0.12	−0.00	−0.00	−0.03	–								
7. Education	5.72	0.72	−0.05	−0.09	0.00	−0.01	−0.18*	0.07	–							
8. Industry – Business services	0.23	0.42	−0.08	−0.08	−0.13	−0.14	−0.15	0.25**	0.01	–						
9. Industry – Management	0.04	0.21	−0.06	−0.09	−0.05	−0.05	0.06	0.00	0.04	−0.12	–					
10. Industry – IT	0.13	0.33	0.14	−0.08	0.05	0.04	0.01	−0.05	−0.06	−0.22**	−0.08	–				
11. Industry – Engineering	0.08	0.28	0.05	0.14	0.05	0.05	0.26**	−0.19*	−0.04	−0.17*	−0.06	−0.12	–			
12. Industry – Medical	0.05	0.23	0.04	0.18*	−0.00	0.04	−0.03	0.01	−0.01	−0.14	−0.05	−0.09	−0.07	–		
13. Knowledge of AI	3.10	0.86	0.07	0.14	0.10	0.12	−0.04	−0.17*	0.03	0.03	−0.02	0.12	−0.01	−0.02	–	
14. ChatGPT usage in the past week	1.89	0.90	0.25**	0.03	0.02	0.07	−0.13	−0.18*	0.04	−0.05	−0.04	0.06	−0.04	0.12	0.23**	–

Note. $N = 174$. 1 = AI condition; 0 = control condition. Gender was dummy coded with 0 = male and 1 = female. Education was coded with 1 = Left high school without a formal qualification; 2 = Completion of compulsory basic primary or secondary schooling; 3 = Completion of GCSEs or equivalent qualifications; 4 = Completion of A-Levels or equivalent qualifications; 5 = Completion of apprenticeship or training; 6 = Technical college or university degree/Ph.D./postdoctoral qualification. Industry variables are dummy coded, with "other" as the reference category. Knowledge about AI was coded with 1 = I have no knowledge; 2 = Novice: I have heard of AI; 3 = Intermediate: I have read media articles or have listened to news about AI technologies; 4 = Advanced: I have used AI-based tools and have some understanding of how they work; 5 = Expert: For example, I am an academic or industry researcher in AI.
* $p < .05$. ** $p < .01$.

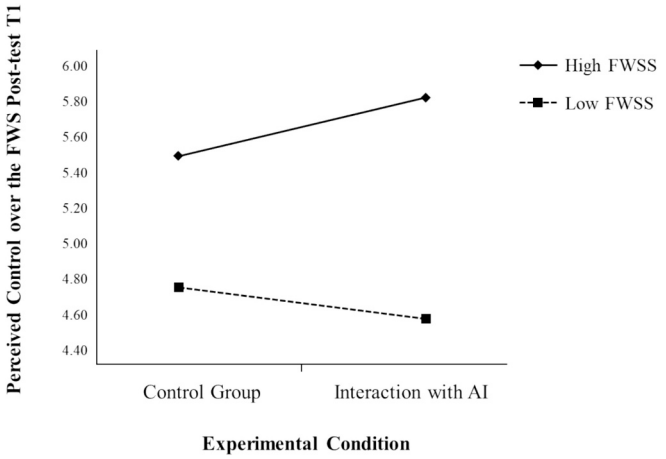


Fig. 3. Study 1: Moderating effect of FWSS on the relationship between experimental condition and perceived control over the FWS T1.

multivariate outliers applying Mahalanobis distance (cutoff *p*-value of 0.001, Tabachnick & Fidell, 2013). Twelve participants were excluded due to failed attention checks, while no exclusions were made based on Mahalanobis distance, leaving us with a final sample of 208 participants (44 % identified as women, the remaining participants identified as men; age: $M = 22.41$ years, $SD = 3.39$).

The procedure for Study 2 mirrored that of Study 1 (see Fig. 2), with a technical modification to the AI-assisted condition. In this study, we integrated ChatGPT directly into our experimental interface via an HTML chat window in Qualtrics, eliminating the need for participants to interact with ChatGPT in a separate browser window. This allowed communication between participants and ChatGPT, enabled by an API call through Javascript.

4.1.2. Measures

4.1.2.1. Future work self salience (pre-test) and perceived control over the future work self (post-test Time 1 and 2). FWSS ($\alpha = 0.82$) and future work self control ($\alpha = 0.80$ and 0.86 for T1 and T2, respectively) were assessed with the same measures as in Study 1.

4.1.2.2. Control variables. Consistent with Study 1, we controlled for age, gender, education (i.e., whether students already had a bachelor's degree or only a high school diploma), and self-reported knowledge about AI technology as possible controls at T1, and ChatGPT usage in the past week at Time 2. Additionally, as participants in this study may have already been familiar with ChatGPT, we included overall ChatGPT usage as a potential control variable ("How often do you use ChatGPT?"; *1 = never, 7 = every day*) because regular ChatGPT users may be less susceptible to our experimental manipulation. The pattern of results remains the same when these covariates are not included (Bernerth & Aguinis, 2016).

4.2. Results

Descriptive statistics and correlations of study variables can be found in Table 2. We again followed Preacher et al. (2007) and Hayes's (2022) recommendations to examine our first hypothesis and employed the SPSS PROCESS macro (Model 1, Hayes, 2022). The moderator (FWSS) was mean-centered before its interaction term was created.

As seen in Table 4 (Study 2), we conducted a similar analysis to Study 1 using FWSS as the moderator, experimental condition as the predictor, and control over future work self T1 (Model 1) and T2 (Model 2) as the dependent variable.

The main effect of the experimental condition on perceived control over the future work self at T1 was not significant ($\beta = 0.14$, $p = .16$; Table 4, Study 2, Model 1). Moreover, the interaction effect between experimental condition and FWSS on perceived control over the future work self at T1 was also not significant ($\beta = 0.12$, $p = .23$; Table 4, Study 2, Model 1).

We conducted the same analysis for control over the future work self at T2. Again, the main effect of the experimental condition on perceived control over the future work self at T2 was not significant ($\beta = 0.14$, $p = .17$; Table 4, Study 2, Model). However, as expected, there was a significant interaction between the experimental condition and FWSS on perceived control over the future work self at T2 ($\beta = 0.26$, $p < .05$; Table 4, Study 2, Model 2). A simple slopes analysis revealed that individuals with high FWSS (1 *SD* above the mean) showed a positive and significant effect of the experimental condition on perceived control over the FWS (slope = 0.37, $p = .01$) compared to those with low FWSS (1 *SD* below the mean; slope = −0.08, $p = .56$; see Fig. 4). The difference between these slopes was significant ($z = 2.27$, $p < .05$).

4.3. Discussion

The findings of Study 2 largely replicate those of Study 1 and support Hypothesis 1, demonstrating a significant positive

Table 2
Descriptive statistics and correlations between study variables (Study 2).

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Condition	0.48	0.50	–														
2. FWSS	3.35	0.88	−0.07	–													
3. Perceived control over the FWS T1	5.52	0.90	0.03	0.25**	–												
4. Perceived control over the FWS T2	5.49	0.90	0.04	0.30**	0.73**	–											
5. Age (years)	22.4	9.94	0.10	0.00	−0.23**	−0.16*	–										
6. Gender	0.44	0.49	−0.05	0.00	−0.05	−0.03	0.08	–									
7. Education	4.49	1.20	0.12	−0.07	−0.14*	−0.10	0.57**	0.11	–								
8. Aspired-to industry – Business services	0.32	0.47	−0.04	−0.05	0.00	0.00	−0.15*	0.08	−0.10	–							
9. Aspired-to industry – Management	0.31	0.46	0.06	−0.01	0.10	0.08	−0.05	−0.09	−0.01	−0.48**	–						
10. Aspired-to industry – IT	0.08	0.28	−0.02	0.03	−0.06	−0.09	0.09	0.07	0.15*	−0.22**	−0.21**	–					
11. Aspired-to industry – Engineering	0.05	0.23	0.05	0.08	0.02	0.04	0.06	−0.09	−0.01	−0.17*	−0.17*	−0.07	–				
12. Aspired-to industry – Medical	0.02	0.16	0.06	0.07	0.02	−0.03	0.02	0.19**	0.02	−0.12	−0.11	−0.05	−0.04	–			
13. Knowledge of AI	3.20	0.78	−0.08	0.04	0.20**	0.10	−0.04	−0.15*	0.07	−0.06	0.10	0.12	−0.03	0.03	–		
14. Overall ChatGPT usage	3.73	1.98	−0.04	−0.02	−0.02	−0.05	−0.08	−0.23**	−0.10	0.04	0.12	0.07	−0.07	−0.12	0.46**	–	
15. ChatGPT usage in the past week	1.88	1.11	0.13	0.06	0.03	−0.03	0.01	−0.29**	−0.08	0.04	0.01	0.20**	−0.03	−0.14*	0.39**	0.60**	–

Note. $N = 208$. 1 = AI condition; 0 = control condition. Gender was dummy coded with 0 = male and 1 = female. Education was coded with 1 = Left high school without a formal qualification; 2 = Completion of compulsory basic primary or secondary schooling; 3 = Completion of GCSEs or equivalent qualifications; 4 = Completion of A-Levels or equivalent qualifications; 5 = Completion of apprenticeship or training; 6 = Technical college or university degree/Ph.D./postdoctoral qualification. Industry variables are dummy coded, with "Other" as the reference category. Knowledge about AI was coded with 1 = I have no knowledge; 2 = Novice: I have heard of AI; 3 = Intermediate: I have read media articles or have listened to news about AI technologies; 4 = Advanced: I have used AI-based tools and have some understanding of how they work; 5 = Expert: For example, I am an academic or industry researcher in AI.
* $p < .05$. ** $p < .01$.

relationship between interacting with an AI and perceived control over the FWS at T2 for participants with high FWSS, but not for those with low FWSS.

We next investigated the full moderated mediation model, including the effect of future work self control on proactive career behavior. Having established that FWSS moderates the impact of interacting with an AI on future work self control for working adults who had never before used ChatGPT, as well as for students with varying levels of ChatGPT usage, we sought to replicate this finding in full-time employees who have already used ChatGPT. Thus, in Study 3, we collected data from full-time employees in Germany with varying levels of ChatGPT use to test our full model.

5. Study 3

5.1. Method

5.1.1. Sample and procedure

Participants were full-time employees in Germany, recruited for study participation by students at a large German business school. One hundred eighty-one full-time employees participated in the first part of the experiment and 169 in the second part. We eliminated three participants whose surveys were incomplete and two participants who could not be matched between the two time points. In line with our approach in Studies 1 and 2, following recent guidelines (Ward & Meade, 2022; Zickar & Keith, 2023), we examined for careless responses with instructed response items (Meade & Craig, 2012) and multivariate outliers with Mahalanobis distance ($p < .001$ at the respective $\chi 2$ value, Tabachnick & Fidell, 2013). This led to the exclusion of eight participants who did not correctly respond to more than one attention check (Meade & Craig, 2012) and one participant who exceeded the threshold for the Mahalanobis distance, leaving a final sample of 155 (44.5 % identified as women, the remaining participants identified as men; age: $M = 32.18$ years, $SD = 9.17$). When the data were analyzed, including this outlier, the pattern of results remained the same.

The procedure for Study 3 was the same as in Study 2, with the addition of proactive career behavior assessed at T2 (see Fig. 2).

5.1.2. Measures

5.1.2.1. Future work self salience (pre-test) and perceived control over the future work self (post-test Time 1). FWSS ($\alpha = 0.89$) and future work self control ($\alpha = 0.88$) were assessed with the same measures as in Study 1.

5.1.2.2. Proactive career behavior (post-test Time 2). Proactive career behavior was assessed with a 13-item scale (Cronbach's $\alpha = 0.92$) used in prior research (e.g., Strauss et al., 2012). The scale includes four sub dimensions: career planning, career consultation, skill development, and networking. Examples of items are: "I am planning what I want to do in the next few years of my career." (career planning); "I make my supervisor aware of my work aspirations and goals." (career consultation); "I develop skills which may not be needed so much now but in future positions." (skill development); "I am building a network of colleagues I can call on for support." (networking). All responses were given on a scale ranging from 1 (not at all) to 5 (a great deal).

5.1.2.3. Control variables (Time 1 and 2). Consistent with Studies 1 and 2, we controlled for age, gender, education, knowledge of AI, and participants' overall ChatGPT usage in all analyses. We found the same pattern of results when these covariates are not included (Bernerth & Aguinis, 2016).

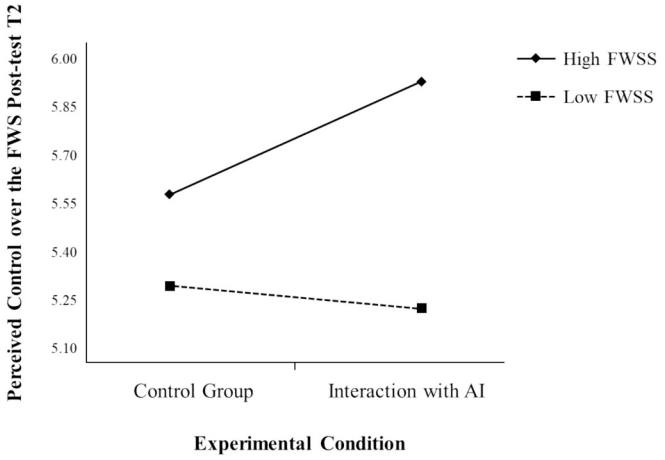


Fig. 4. Study 2: Moderating effect of FWSS on the relationship between experimental condition and perceived control over the FWS T2.

5.2. Results

Descriptive statistics and intercorrelations of study variables can be found in Table 3. To examine our complete model (see Fig. 1), we employed first-stage moderated mediation analyses (Preacher et al., 2007) using the SPSS PROCESS macro (Model 7; Hayes, 2022). Again, the moderator (FWSS) was mean-centered before its interaction term was created.

The main effect of the experimental condition on perceived control over the future work self at T1 was not significant ($\beta = -0.01$, $p = .94$; Table 4, Study 3, Model 1). Supporting Hypothesis 1, the interaction between experimental condition and FWSS on perceived control over the future work self at T1 was again significant ($\beta = 0.44$, $p < .01$; Table 4, Study 3, Model 1). A simple slopes analysis revealed that for individuals with high FWSS (1 *SD* above the mean) there was a positive and significant effect of the experimental condition on perceived control over the future work self (slope = 0.45, $p < .05$). However, for those with low FWSS there was a negative and significant effect of the experimental condition on perceived control over the future work self (1 *SD* below the mean; slope = −0.47, $p < .05$). The difference between these slopes was significant ($z = 3.40$, $p < .01$; see Fig. 5).

To test Hypothesis 2, we conducted moderated mediation analyses (Preacher et al., 2007) using SPSS version 26.0 (IBM Corp., 2019) and the PROCESS macro (Model 7; Hayes, 2022). We examined our hypotheses using bootstrapping with 5000 resamples (Preacher & Hayes, 2004) and 95 % confidence intervals. In line with our hypothesis, perceived control over the future work self at T1 was positively related to proactive career behavior at T2 ($\beta = 0.42$, $p < .001$; Table 4, Study 3, Model 2), and FWSS moderated the indirect effect between experimental condition and proactive career behavior at T2 through control over future work self at T1 (index of moderated mediation = 0.20, 95 % CI [0.07; 0.34]). Specifically, the conditional indirect effect of experimental condition on proactive career behavior at T2 via control over the future work self at T1 was positive when FWSS was high (1 *SD* above the mean) (indirect effect = 0.20, 95 % CI [0.04; 0.36]), but negative when FWSS was low (1 *SD* below the mean) (estimate = −0.23, 95 % CI [−0.43; −0.02]). This difference was statistically significant (contrast = 0.43, 95 % CI [0.15; 0.73). These results fully support Hypothesis 2.

6. General discussion

Although rapid advances in AI are poised to transform the world of work and impact individuals' career-related cognition and behavior, the ways in which individual differences shape these responses remain poorly understood. Using the proactive motivation model as our overarching theoretical framework, we developed a model of how future work selves shape individuals' reactions to interacting with a powerful generative AI. Across three studies, we found that FWSS shapes the effect of interacting with an AI on individuals' future work self control. Specifically, the relationship between interacting with an AI (versus the control condition) and future work self control was positive for participants with high FWSS (Studies 1, 2, and 3). Interacting with an AI seemed to boost individuals' sense of control over their future work self (a "can do" motivational factor), but only when they had a clear vision of their future work self (a "reason-to" motivational factor (S. K. Parker et al., 2010)). For participants with low FWSS, the results were somewhat less consistent. Interacting with an AI did not affect their perceived control over their future work self in Studies 1 and 2. However, in Study 3, participants low in FWSS felt less in control of their future work self after interacting with an AI. Study 3 further extended the findings of Studies 1 and 2 by providing evidence for our proposed moderated mediation model. Specifically, the indirect relationship between interacting with an AI and proactive career behavior via future work self control was moderated by FWSS such that the indirect effect was positive when individuals' FWSS was high and negative when individuals' FWSS was low. Participants who interacted with an AI thus reported higher levels of proactive work behavior when their future work self was salient, but lower levels when it was not.

7. Theoretical implications

Our paper highlights future work selves as a useful theoretical lens to explore the impact of AI on individuals' careers. Our results reconcile contradictory findings on individuals' reactions to artificial intelligence. Specifically, we identify FWSS as an important individual difference that determines whether AI is seen as a threat, taking control over their future career away from individuals, or as an opportunity, increasing their sense of control.

We contribute to burgeoning research on the impact of AI on individuals' career-related cognition and behavior by adopting a future-oriented perspective. To date, studies of employees' views of AI have predominantly taken a"present" perspective (e.g., Bochniarz et al., 2022; Kong et al., 2023; Langer et al., 2023), for instance, by investigating their perceptions of the impact of AI on their current employability (Bhargava et al., 2021). Yet the concept of prospective sensemaking (Gioia et al., 1994; Gioia & Thomas, 1996) implies that reactions to current experiences, like AI interactions, are also strongly shaped by individuals' conceptions of what this might mean for them in the future. By highlighting the stabilizing effects of a salient future work self, our research shows that individuals interpret AI interactions not only in terms of immediate challenges and opportunities, but also in terms of how these interactions align with and inform their future careers. We thus add a new perspective to recent findings (Tang et al., 2023) that interactions with AI can yield both positive and negative outcomes. By introducing FWSS as an important individual difference, we illuminate why some individuals perceive AI as an ally in shaping their future, while others view it as a challenge to their career aspirations. A clear and easily accessible image of who they want to become in their future career may allow individuals interacting with an AI to mentally simulate possible future scenarios that enable them to see AI as an opportunity rather than a threat.

Drawing on social psychological research on the broader concept of possible selves (Norman & Aron, 2003; Oyserman & James, 2009), our research extends the literature on future work selves by focusing not only on their salience, the extent to which a future

Table 3
Descriptive statistics and correlations between study variables (Study 3).

Variable	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1. Condition	0.49	0.50	–														
2. FWSS	3.47	1.04	−0.06	–													
3. Perceived control over the FWS T1	5.46	0.95	0.00	0.39**	–												
4. Proactive Career Behavior T2	3.35	0.79	0.00	0.37**	0.51**	–											
5. Age (years)	32.2	9.16	−0.06	0.19*	−0.14	−0.15	–										
6. Gender	0.45	0.49	−0.10	−0.00	−0.11	0.02	−0.03	–									
7. Education	5.72	0.72	0.11	−0.07	0.03	0.12	−0.10	−0.01	–								
8. Industry – Business services	0.23	0.42	−0.06	−0.11	−0.11	−0.05	0.00	0.04	−0.08	–							
9. Industry – Management	0.21	0.41	−0.07	0.05	0.12	0.11	0.01	−0.05	0.02	−0.29**	–						
10. Industry – IT	0.08	0.27	0.17*	−0.21**	−0.02	−0.05	−0.02	−0.18*	0.02	−0.17*	−0.15	–					
11. Industry – Engineering	0.07	0.25	0.08	−0.01	0.02	−0.05	−0.03	−0.04	0.00	−0.15	−0.14	−0.08	–				
12. Industry – Medical	0.11	0.32	0.05	0.03	−0.03	−0.04	−0.06	0.16*	0.00	−0.20*	−0.19*	−0.11	−0.10	–			
13. Knowledge of AI	3.10	0.86	0.03	0.04	0.25**	0.11	−0.20*	−0.16*	0.08	−0.11	0.04	0.24**	0.17*	−0.20*	–		
14. Overall ChatGPT usage	3.25	1.84	−0.00	0.00	0.11	0.12	−0.31**	−0.22**	0.16*	0.14	0.04	0.06	0.13	−0.18*	0.43**	–	
15. ChatGPT usage in the past week	1.89	0.90	0.03	−0.02	0.12	0.06	−0.26**	−0.27**	0.06	0.20*	−0.04	0.08	0.14	−0.16*	0.23**	0.66**	–

Note. $N = 155$. Condition was coded with 1 = AI condition and 0 = control condition. Gender was dummy coded with 0 = male and 1 = female. Education was coded with 1 = Left high school without a formal qualification; 2 = Completion of compulsory basic primary or secondary schooling; 3 = Completion of GCSEs or equivalent qualifications; 4 = Completion of A-Levels or equivalent qualifications; 5 = Completion of apprenticeship or training; 6 = Technical college or university degree/Ph.D./postdoctoral qualification. Industry variables are dummy coded, with "Other" as the reference category. Knowledge about AI was coded with 1 = I have no knowledge; 2 = Novice: I have heard of AI; 3 = Intermediate: I have read media articles or have listened to news about AI technologies; 4 = Advanced: I have used AI-based tools and have some understanding of how they work; 5 = Expert: For example, I am an academic or industry researcher in AI.
* $p < .05$. ** $p < .01$.

Table 4
Regression results with interaction terms (Studies 1, 2, and 3).

Dependent variable	Perceived control over the FWS										Proactive career behavior	
	Study 1				Study 2				Study 3			
	Model 1		Model 2		Model 1		Model 2		Model 1		Model 2	
Predictors	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*
Age	−0.00	0.01	−0.00	0.01	−0.04*	0.02	−0.03	0.02	−0.02**	0.01	−0.01	0.01
Gender	0.14	0.12	0.14	0.13	−0.05	0.10	−0.05	0.11	−0.20	0.14	0.12	0.12
Education	0.04	0.05	0.02	0.05	−0.03	0.05	−0.02	0.05	0.03	0.10	0.10	0.08
Knowledge of AI	0.05	0.08	0.05	0.09	0.25**	0.07	0.16*	0.08	0.28**	0.09	−0.06	0.07
Overall ChatGPT usage					−0.06*	0.03	−0.05	0.03	−0.04	0.04	0.03	0.04
ChatGPT usage in the past week			0.09	0.08								
Condition	0.09	0.12	−0.02	0.12	0.14	0.10	0.14	0.11	−0.01	0.14	−0.00	0.11
Perceived control over the FWS											0.42**	0.06
FWSS	0.39**	0.08	0.39**	0.08	0.13	0.08	0.14	0.08	0.14	0.10		
Condition x FWSS	0.26*	0.12	0.25*	0.12	0.13	0.11	0.26*	0.12	0.44**	0.13		
R^2	0.32**		0.32**		0.18**		0.16**		0.30**		0.28**	
ΔR^2	0.02*		0.02*		0.01		0.02*		0.05**			

Note. $N_{Study\ 1} = 174$. $N_{Study\ 2} = 208$. $N_{Study\ 3} = 155$. Condition was coded with 1 = AI condition and 0 = control condition. Gender was dummy coded with 0 = male and 1 = female. Education was coded with 1 = Left high school without a formal qualification; 2 = Completion of compulsory basic primary or secondary schooling; 3 = Completion of GCSEs or equivalent qualifications; 4 = Completion of A-Levels or equivalent qualifications; 5 = Completion of apprenticeship or training; 6 = Technical college or university degree/Ph.D./postdoctoral qualification. Knowledge about AI was coded with 1 = I have no knowledge; 2 = Novice: I have heard of AI; 3 = Intermediate: I have read media articles or have listened to news about AI technologies; 4 = Advanced: I have used AI-based tools and have some understanding of how they work; 5 = Expert: For example, I am an academic or industry researcher in AI.
* $p < .05$. ** $p < .01$. *** $p < .001$.

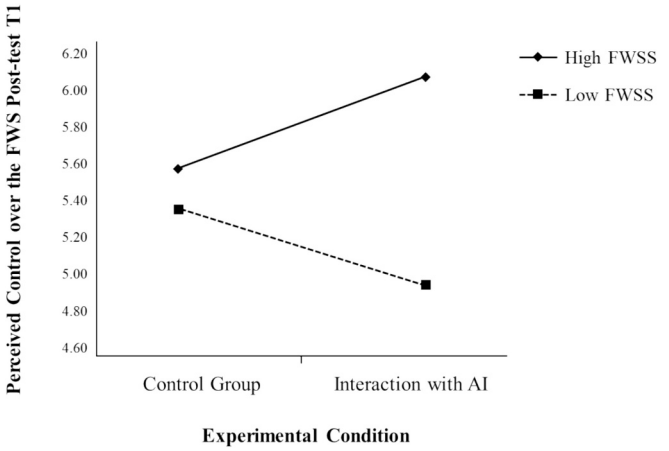


Fig. 5. Study 3: Moderating effect of FWSS on the relationship between experimental condition and perceived control over the FWS T1.

work self is easily accessible and clear (Strauss et al., 2012), but also on the perceived control over the future work self. As suggested by the proactive motivation model (S. K. Parker et al., 2010), salience and perceived control interact in predicting proactive career behavior, reflecting reason-to and can-do factors, respectively.

While our validation study demonstrated that individuals' can-do motivation in relation to their future work self, future work self control, is distinct from other control related concepts in the careers literature, such as career confidence or career-related self-efficacy, our findings contribute to this wider body of research, which highlights the centrality of control beliefs in relation to career management. For example, Kossek et al. (1998) demonstrated that individuals' beliefs about their ability to control career-related outcomes are a key mechanism translating contextual factors into career self-management behavior. Similarly, social cognitive career theory suggests that contextual supports and barriers shape individuals' career-related control beliefs, which in turn determine their career-oriented actions and outcomes (Lent & Brown, 2013). In our theoretical model, AI, which can be seen as both a support or a barrier to individuals' careers, either positively or negatively affects their career-related control beliefs, depending on future work self salience. Future work self salience is thus an individual difference that moderates the influence of context on individuals' control beliefs in relation to their future career. In the face of uncertainty caused by interaction with a powerful AI, a salient future work self seems to act as a personal resource that allows individuals to maintain a sense of control over their desired future career. In addition to

providing support for the interaction of can-do and reason-to motivational factors proposed by the proactive motivation model (S. K. Parker et al., 2010), the findings of Study 3 also align with the conceptual framework of whole-life career self-management (Hirschi et al., 2020) which similarly argues that the interplay between the valence of a goal (in our case, a salient vision of a highly valued future work self) and expectancy (in our case, the perceived control over the future work self) shapes individuals' career-related action regulation.

8. Limitations and future research

While our studies have a number of strengths, they also have limitations. First, even though we find consistent support for the buffering effect of high FWSS, in Studies 1 and 2, participants with *low* FWSS did not perceive less control over their FWS after AI interaction. Yet, in Study 3, participants with low FWSS reported lower future work self control after interacting with the AI, as we had predicted. It may be that the differing characteristics of our three samples account for these differences. For example, the lack of experience with ChatGPT of participants in Study 1 and the limited professional experience of participants in Study 2 may have made it less likely for the absence of a clear vision of their future work self to have a detrimental effect. From a theoretical perspective, this may suggest that experience with AI and with managing one's career may be potential moderators of the relationships we proposed. Missing out on the benefits of mental simulation of the future may be particularly detrimental when participants also have little past experience to draw on. Future research is needed to explore this possibility and replicate our finding regarding the negative impact of low FWSS in other samples.

Likewise, it is plausible that confounding factors such as the field of work may impact both individuals' future work self and the extent to which their job is vulnerable to being replaced by technology (Balsmeier & Woerter, 2019; Michaels et al., 2014). Because participants were randomly allocated to conditions in our experimental design, individual differences are unlikely to account for differences in future work self control and proactive career behavior. Yet, to account for the possibility that the interaction with ChatGPT may have seemed less relevant to some participants' daily work, we included three items adapted from Kelly, Strauss, Arnold, and Stride (2020) to measure the similarity between the experimental task and individuals' daily work as a control variable. The inclusion of this control variable did not change the overall pattern of results across the three studies, reinforcing the robustness of our findings and supporting the argument that our results are not biased by differences in task relevance across job types (see Online Supplemental Material for the results of these post-hoc analyses). In addition, we found no significant correlations between dummy variables representing the five most common industries our participants worked in and our dependent variables in Studies 1–3 (see Tables 1–3). Furthermore, multivariate analyses of variance revealed no significant effects of industry on our dependent variables across the three studies, suggesting that industry does not significantly affect the dependent variables (see Online Supplemental Material for the results of these post-hoc analyses). Nevertheless, future research may explore whether our findings generalize to professional settings that are particularly impacted by the rise of AI.

It is also plausible that interacting with ChatGPT could trigger AI aversion, leading participants to perceive AI as a threat (e.g., Dietvorst, Simmons, & Massey, 2015). To account for this possibility we measured participants' attitudes towards AI with two items developed by Gaube et al. (2021) in Studies 2 and 3. We found no differences between control and experimental conditions in either study.[3] We also included attitudes towards AI as an additional control variable in our analyses, but did not find that this changed the pattern of results (see Online Supplemental Material for the results of the post-hoc analyses). This finding aligns with recent studies suggesting that aversion may not be a universal response to interactions with generative AI (Böhm et al., 2023; Chu & Liu, 2023), and suggest that it was perceived control over the future work self, rather than a negative attitude towards the AI, that accounted for our results.

Another possible limitation is that we did not include a manipulation check in our studies. While manipulation checks are necessary when experimental manipulations aim to alter participants' affect or psychological states, "researchers should not run manipulation checks when the experimental treatment consists of objective variations of parameters" (Lonati et al., 2018, p. 22). In our case, interaction with an AI as part of the experimental task was a parameter that varied objectively between conditions. Nevertheless, we sought to establish that our experimental manipulation was indeed effective. We conducted an additional study with 119 full-time employees in the United Kingdom, recruited through Prolific, using the same experimental manipulation as in Studies 1–3. The results provided strong support for the effectiveness of our experimental manipulation[4] (see Online Supplemental Material for details).

A further possible limitation may be that participants in the AI condition could have felt less in control because by providing prompts, the experiment limited their autonomy in interacting with the AI and potentially their sense of control over how they would use the AI in their future careers. We provided prompts in the early stages of the experimental task because, particularly in Study 1, we recruited participants who were reportedly using ChatGPT for the first time during our experiment. Because prompting is challenging for novices (Bašić et al., 2023) and often involves a trial-and-error process even for experts (Zamfirescu-Pereira et al., 2023), we first

[3] Study 2: Control condition: $M = 4.24$, $SD = 1.28$; experimental condition: $M = 4.24$, $SD = 1.28$; $t(206) = -0.02$, $p = .98$. Study 3: Control condition: $M = 4.27$, $SD = 1.17$; experimental condition: $M = 4.15$, $SD = 0.98$; $t(153) = 0.73$, $p = .46$.

[4] Participants were asked to rate the following two statements adapted from Man Tang et al. (2022) on a 7-point Likert scale, ranging from 1 = "Strongly Disagree" to 7 = "Strongly Agree": "I worked on this in-tray task independently" (reverse coded) and "I worked on this in-tray task with the help of ChatGPT3" ($\alpha = 0.80$) after the experimental manipulation. Responses to the manipulation check were significantly different between participants in the experimental group ($M = 5.00$, $SD = 1.17$) and the control group ($M = 1.22$, $SD = 0.67$), $t(117) = -21.46$, $p < .001$, $d = -3.93$, indicating that participants could clearly identify whether or not they had interacted with an AI in completing the task.

showed them how to use prompts before giving them the opportunity to write their own prompts. However, there were no significant differences in future work self control between the experimental and control conditions in any of the three studies, which speaks against this possibility (see Tables 1–3).

Moreover, the exclusive use of self-report measures may introduce potential limitations to our research due to common method bias (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). However, due to three key factors, common method variance (CMV) is less likely to impact our findings. First, we used an experimental design, where participants were randomly allocated to a condition, which is an effective method of eliminating CMV (Antonakis et al., 2010; for a review, see Cooper et al., 2020). Second, we measured our outcome variable at two time points in Studies 1 and 2 and separated the measurement of our mediator and outcome in Study 3. This reduces the effects of common method variance (CMV) (Johnson et al., 2012). Finally, the relationships in our study are dependent on a moderator. Research by Siemsen et al. (2009) demonstrates that interactive relationships cannot be inflated by CMV. Nevertheless, we encourage future research to employ rigorous longitudinal designs such as experience sampling methodologies (ESM) that allow for multiple data collection points (Cooper et al., 2020), e.g., to shed light on how continuous interaction with AI influences FWSS, future work self control, and proactive career behaviors over time.

Additionally, while we have focused on perceived control over one's future work self in addition to its salience, future research may investigate how the content of future work selves changes in the face of rapidly evolving AI landscapes. Consider a radiologist who invested in years of specialized training with a vivid and salient image of what their future will entail. As AI technologies increasingly assume diagnostic tasks, their clear vision might be at odds with the dynamic realities of the AI-driven job market (Brynjolfsson & Mitchell, 2017). Indeed, while a salient future work self can be a driving force, rigidly clinging to such a future work self could become detrimental. While research generally supports the positive relationship between FWSS and career adaptability (e.g., Guan et al., 2014), a highly salient and clearly defined future work self may also have drawbacks. Individuals whose sense of self is closely tied to their work are likely to be more willing to make personal sacrifices in the pursuit of their future, suggesting that they may more vulnerable to exploitation (Bunderson & Thompson, 2009). Similarly, high FWSS is associated with a more focused job search (Fang & Saks, 2022), which may, in some cases, facilitate "tunnel vision" and thus limit the range of labor market opportunities that some individuals consider (Okay-Somerville & Scholarios, 2022). In contrast, a less defined future work self may provide a protective flexibility that allows individuals to remain open to a wider range of career possibilities and adaptations in the face of uncertainty. By not being overly committed to a specific vision of the future, individuals may be better positioned to pivot and explore different opportunities that arise in a rapidly changing work environment. Future research could thus investigate the interplay between the salience of a future work self and its flexibility, offering insights into how individuals can adeptly reconcile their career aspirations with the rapidly evolving work environment, ensuring both career progression and personal well-being.

Last, a possible limitation is that the majority of the employees sampled in Studies 1 and 3 were relatively young. Their future work selves may have been more salient to these comparatively young samples than they would have been to older workers. While this makes them appropriate samples to explore our research question, future research is needed to examine whether these findings hold true for older employees (e.g., those nearing retirement).

9. Practical implications

Our findings hint at a potentially dangerous downward spiral where those who do not have a clear sense of who they would like to be in their future career are more vulnerable to the detrimental effects of AI on their sense of control in relation to their future career, making it in turn less likely for them to develop skills and networks that would allow them to prepare for an uncertain future. Importantly, our findings also imply that a clear picture of one's future work self allows individuals to envision the benefits of AI for their future work, leaving them feeling more in control and thus promoting their proactive career behavior. Hence, our study equips organizations with insights into why certain employees might struggle with AI integration. Consequently, organizations can take proactive steps by integrating deliberate career development endeavors, such as career counseling or mentoring, to foster employees' future work self salience (Kao et al., 2020; Strauss et al., 2012). For example, organizations could expose employees to focused interventions, like mental imagery exercises (Blouin-Hudon & Pychyl, 2017), that can assist employees in creating vivid representations of their future work selves (Strauss & Parker, 2018). In this context, as the majority of jobs will be impacted by AI in some way, individuals should be encouraged to visualize how AI might present opportunities to advance or explore their own careers (Cai et al., 2015).

Moreover, leaders likely have an essential role in helping their employees to develop a clear picture of their professional future. Leaders, as key influencers of employees' self-concept (Avolio et al., 2004), can promote FWSS through vision communication, which enables employees to imagine themselves in the future (Guo et al., 2022).

Finally, given that the participants in Study 2 were university students, our findings highlight the critical role of higher education in preparing students for the AI-impacted job market (Kshetri, 2024). Like organizations, universities should adapt proactive strategies such as career counseling (van der Horst et al., 2021) and potentially integrate mental imagery exercises (Strauss & Parker, 2018) to enhance students' FWSS. In addition, educators, similar to business leaders, can use their influence to inspire and shape students' visions of their future careers. Finally, governments could play a critical role in providing career-related resources and support to those who may not have access to organizational or university programs, ensuring broader access to career guidance and skills development in the evolving AI landscape.

10. Conclusion

We are in the midst of an AI revolution that is fundamentally changing the way we work. It is thus critical to understand how individual differences influence people's perceptions of the impact of AI on their future career. Our research highlights the role of FWSS in moderating the impact of AI on individuals' perceptions of control over their professional future. Individuals who interact with AI and have a salient future work self experienced a greater sense of control over their future career, which encourages proactive career behavior. In contrast, individuals with a less salient future work self experienced a decrease in control and exhibit less proactive career behavior, at least in some circumstances. As the integration of AI becomes more widespread, this research provides important insights into the role of AI in shaping the future-oriented career-related cognitions and behaviors of individuals.

CRediT authorship contribution statement

Julian Voigt: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Karoline Strauss:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I've shared the link to our data in the manuscript and on the title page.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jvb.2024.104054.

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